

Tejas Patel

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EDUCATION

Sardar Patel Institute of Technology

Bachelor of Technology in Computer Science and Engineering (Data Science) - 8.44 CGPA

Mumbai

Expected 2026

Ramnarain Ruia College

Higher Secondary Certificate - 87%

Mumbai

2022

PROJECTS

Text to Video Generator | Python, MoviePy, FFmpeg, HTML, JavaScript

[Extension](#)

- Designed and implemented a user-friendly **Chrome extension** that allows users to input text and generate a video, integrating **audio narration**, **image generation**, and **video composition** for an engaging multimedia experience. Utilized **Chrome's storage API** to save and restore generation states, video data, and logs, ensuring continuity across sessions and robust error recovery.
- Created a **Flask backend** with **RESTful APIs** for audio(**ElevenLabs**, **Murf**), image generation(**Gemini**), and timing estimation(**Assembly AI**).
- Deployed on **AWS EC2** with **Supervisor** and **swap memory** for enhanced stability under load.
- Used **ThreadPoolExecutor** to cut image generation time by **50%**.

Content Moderation System | React.JS, Tailwind CSS, MongoDB, Express, Docker

[Github](#)

- Developed a **distributed content moderation system** using **Node.js**, **React**, and **MongoDB**, enabling real-time content processing with three parallel workers decoupled via **RabbitMQ**.
- Designed a **Dockerized architecture** with **NGINX**, **MongoDB**, and **RabbitMQ**, ensuring scalable deployment and high availability. Implemented **worker heartbeat monitoring**, updating statuses every 10 seconds for enhanced system reliability.
- Built a **React dashboard** to display worker status and moderation metrics, integrating **Perspective API** for automated content analysis. Demonstrated fault tolerance and scalability by simulating worker failures, optimizing system performance with **RESTful APIs**.

System Design Synthesis | Agentic Architecture, LLM, Next.js, Express.js, CrewAI, Locust

[Github](#)

- Developed a **multi-agent pipeline** (System Architect, Diagrammer, Reviewer) orchestrated via **CrewAI**, utilizing a **fine-tuned TinyLlama 1.5 model** from **Hugging Face**, trained specifically for architectural reasoning tasks, with fine-tuning using **QLORA** and **AdamW optimizer** to enhance contextual understanding of **system design requirements**.
- Generated a full system design proposal, including architecture pattern, core components, technology stack, data flow, rationale, and **PlantUML diagrams**.
- Produced basic **code base**, a **docker-compose.yml** file to run the system, and a **Locust** server for load testing.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript, HTML/CSS

Databases: MySQL, MongoDB

Frameworks: React.JS, NextJS

Developer Tools: Github, Postman, Docker

Libraries: Pandas, NumPy, Matplotlib, Seaborn, ScikitLearn, TensorFlow, Pytorch, OpenCV

HONORS AND AWARDS

NIT Silchar Hackathon: 1st prize (Open Category)

Frontend Odyssey: 1st runner up

FLAME AI Challenge 2024: Top 40 (Kaggle Competition)

Hactoberfest 2024: [Level 4](#)

EXTRA-CURRICULAR

Head of Marketing, Oculus (Funzone and Esports): Led a team of six members and raised funds from various sponsorship and collaboration deals for college's biggest technocultural fest.

Sangeet Visharad: 1st class certified from Akhil Bhartiya Gandharva Mahavidyalaya in Keyboard (Hindustani Classical Music)